The Unit Circle and Exact Values

Exact values

1 Write the exact value of each expression:

a $\sin\frac{\pi}{6}$ **b** $\cos\frac{3\pi}{4}$ **c** $\tan\frac{5\pi}{6}$ **d** $\sin\frac{3\pi}{2}$

2 Write the exact value of each expression:

a $\cos\left(-\frac{\pi}{3}\right)$ **b** $\sin\frac{7\pi}{4}$ **c** $\tan\pi$ **d** $\cos\frac{11\pi}{6}$

Points on the unit circle

3 The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin\theta$ **b** $\cos\theta$

c $\tan\theta$ **d** the quadrant containing θ

4 State the reference angle for each angle:

a $\frac{5\pi}{6}$ **b** $\frac{4\pi}{3}$ **c** $-\frac{\pi}{4}$ **d** $\frac{7\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

a $\sin\theta < 0$ and $\cos\theta > 0$

b $\tan\theta > 0$ and $\sin\theta < 0$